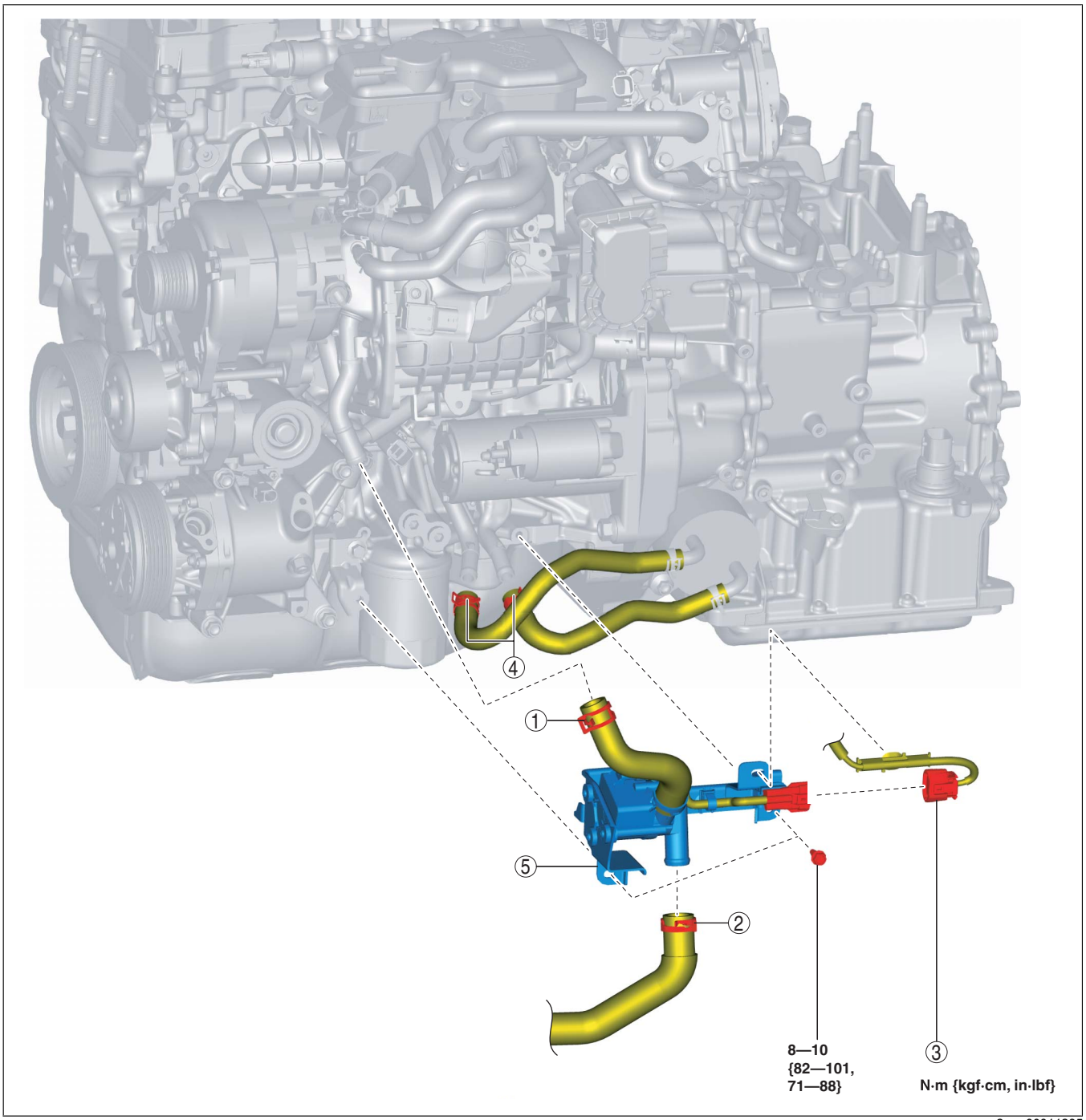
## ELECTRIC WATER PUMP REMOVAL/INSTALLATION [SKYACTIV-D 1.5]

id0112q2702000

**Warning**

- **If the cooling system cap or the water-cooled charge air cooler reserve tank cap is removed when the engine is hot, steam and boiling water may be injected, causing severe burns or severe injury. Perform the removal of the cooling system cap and water-cooled charge air cooler reserve tank cap when the engine is cool.**
- **Turn off the engine and wait until it is cool. Even then, be very careful when removing the cap. Wrap a thick cloth around it and slowly turn it counterclockwise to the first stop. Step back while the pressure escapes.**
- **When you are sure all the pressure is gone, press down on the cap using the cloth, turn it, and remove it.**

1. Disconnect the negative battery cable. (See NEGATIVE BATTERY CABLE DISCONNECTION/ CONNECTION.)
2. Remove the engine cover. (See ENGINE COVER REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
3. Remove the front under cover No.1 and No.2. (See FRONT UNDER COVER No.1 REMOVAL/ INSTALLATION.)(See FRONT UNDER COVER No.2 REMOVAL/INSTALLATION.)
4. Drain the following coolant. (See ENGINE COOLANT REPLACEMENT [SKYACTIV-D 1.5].)(See WATER-COOLED CHARGE AIR COOLER COOLANT REPLACEMENT [SKYACTIV-D 1.5].)
   - ATX: Engine coolant and water-cooled charge air cooler coolant
   - MTX: Water-cooled charge air cooler coolant
5. Remove the coolant reserve tank. (See COOLANT RESERVE TANK REMOVAL/INSTALLATION [SKYACTIV-D 1.5].)
6. Remove in the order indicated in the table.
7. Install in the reverse order of removal.
8. Refill the following coolant. (See ENGINE COOLANT REPLACEMENT [SKYACTIV-D 1.5].)(See WATER-COOLED CHARGE AIR COOLER COOLANT REPLACEMENT [SKYACTIV-D 1.5].)
   - ATX: Engine coolant and water-cooled charge air cooler coolant
   - MTX: Water-cooled charge air cooler coolant
9. Inspect the following coolant leakage. (See ENGINE COOLANT LEAKAGE INSPECTION [SKYACTIV-D 1.5].) (See WATER-COOLED CHARGE AIR COOLER COOLANT LEAKAGE INSPECTION [SKYACTIV-D 1.5].)
   - ATX: Engine coolant and water-cooled charge air cooler coolant
   - MTX: Water-cooled charge air cooler coolant



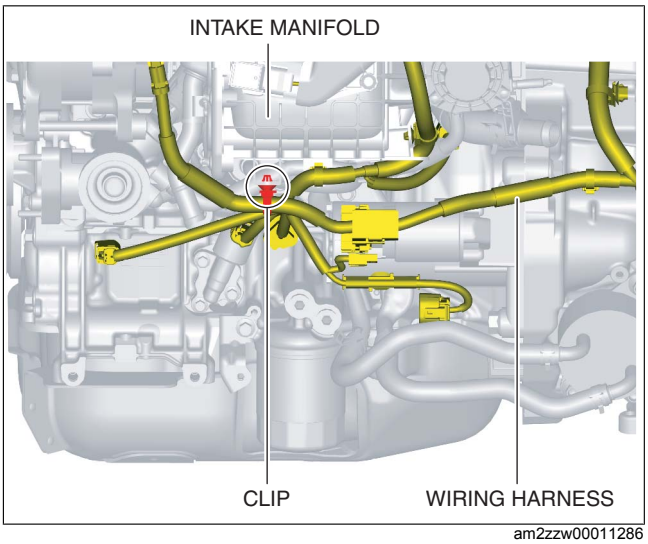

am2zzw00011285

| | |
|---|---|
| 1 | Water hose (water-cooled charge air cooler side) (See Water Hose (water-cooled charge air cooler side) Removal Note.) |
| 2 | Water hose (electric water pump side) |
| 3 | Electric water pump connector, wiring harness |
| 4 | ATF oil cooler water hose (engine side) (ATX) |
| 5 | Electric water pump (See Electric Water Pump Removal Note.) |

**Water Hose (water-cooled charge air cooler side) Removal Note**

1. Disconnect the service plug. (with i-ELOOP) (See SERVICE PLUG DISCONNECTION/CONNECTION [i-ELOOP].)

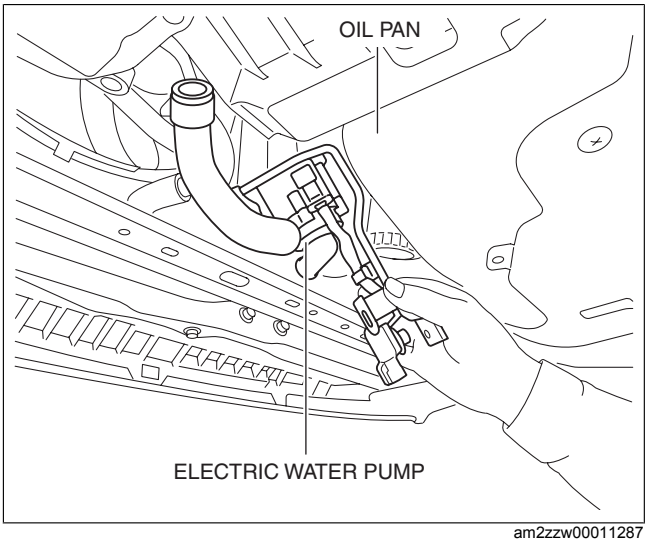
2. Detach the wiring harness clip shown in the figure from the intake manifold.
3. Disconnect generator terminal B and set the wiring harness aside. (See GENERATOR REMOVAL/ INSTALLATION [WITH i-ELOOP (SKYACTIV-D 1.5)].) (See GENERATOR REMOVAL/ INSTALLATION [WITHOUT i-ELOOP (SKYACTIV-D 1.5)].)
4. Disconnect the water hose.



am2zzw00011286

**Electric Water Pump Removal Note**

1. Remove the electric water pump from the underside of the vehicle.



am2zzw00011287